

Ziyang Wei

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Education

Ph.D. in Statistics Department of Statistics, University of Chicago, Chicago, USA	2022 – now
M.Sc. in Statistics Department of Statistics, University of Chicago, Chicago, USA	2020 – 2022
B.Sc. in Mathematics and Applied Mathematics School of Mathematical Sciences, Fudan University, Shanghai, China	2016 – 2020

Research

9. **Ziyang Wei**, Jiaqi Li, Lan Wang, and Wei Biao Wu. [Finite-sample distribution theory and efficient large-scale inference for online quantile regression](#). *Journal of the American Statistical Association (Theory and Methods)*, under review. (**Job Market Paper**)
8. Xiaoli Li, Qianqian Lei, **Ziyang Wei**, and Wei Biao Wu. Exact tail laws for nonlinear SGD with infinite-variance gradients. *International Conference on Learning Representations*, under review.
7. **Ziyang Wei**, Jiaqi Li, Likai Chen, and Wei Biao Wu. [Central limit theorems for stochastic gradient descent quantile estimators](#). *IEEE Transactions on Information Theory*, 2026.
6. **Ziyang Wei**, Wanrong Zhu, Jingyang Lyu, and Wei Biao Wu. [Refining covariance matrix estimation in stochastic gradient descent through bias reduction](#). *International Conference on Artificial Intelligence and Statistics*, 2026.
5. **Ziyang Wei**, Wanrong Zhu, and Wei Biao Wu. [General weighted averaging in stochastic gradient descent: CLT and adaptive optimality](#). *International Conference on Artificial Intelligence and Statistics*, 2026.
4. **Ziyang Wei**, Jiaqi Li, Zhipeng Lou, and Wei Biao Wu. [Gaussian approximation and concentration of constant learning-rate stochastic gradient descent](#). *Advances in Neural Information Processing Systems*, 2025.
3. Or Goldreich, **Ziyang Wei**, Soham Bonnerjee, Jiaqi Li, and Wei Biao Wu. [Asymptotic theory of SGD with a general learning-rate](#). *Advances in Neural Information Processing Systems*, 2025.
2. Wanrong Zhu, Zhipeng Lou, **Ziyang Wei**, and Wei Biao Wu. [High confidence level inference is almost free using parallel stochastic optimization](#). *Journal of Machine Learning Research*, revision submitted.
1. **Ziyang Wei** and Shuqin Zhang. [CALLR: a semi-supervised cell-type annotation method for single-cell RNA sequencing data](#). *Bioinformatics*, 2021. (Oral presentation at ISMB/ECCB 2021)

Teaching

Teaching Assistant

- STAT 25100: Introduction to Mathematical Probability, Winter 2023, Winter 2026
- STAT 23400: Statistical Models and Methods, Spring 2023, Autumn 2023, Autumn 2024
- STAT 22200: Linear Models and Experimental Design, Spring 2024
- STAT 32950/24620, FINM 34700: Multivariate Statistical Analysis: Applications and Techniques, Spring 2025
- STAT 25150: Introduction to Mathematical Probability-A, Spring 2026
- STAT 22000: Statistical Methods and Applications, Autumn 2026

Awards

Chinese Government Award for Outstanding Self-financed Students Abroad

China Scholarship Council, 2025 – 2026

Best Statistical Consulting Team Award

Department of Statistics, University of Chicago, 2024 – 2025

Ph.D. Program Fellowship

University of Chicago, 2022 – 2027

Merit-Based Scholarship for Graduate Students

University of Chicago, 2020 – 2022

Outstanding Graduate

Fudan University, 2020

Top-notch Talent Program Honored Graduate

Fudan University, 2020

Outstanding Student

Fudan University, 2018 – 2019

Experience

Statistical Consultant

2020 – 2025

Department of Statistics, University of Chicago, Chicago, USA

Summer Research Intern

2019

Department of Biostatistics, Harvard T.H. Chan School of Public Health, Boston, USA